

Skeleton Project Report

1. Reading this week – Journal Log Interview

Momin et al. (2025) present a comprehensive survey of explainable deepfake detection systems across visual, audio, and textual domains. The study highlights that although current detectors achieve high accuracy, most remain opaque and difficult to trust. The authors categorise explainability techniques such as Grad-CAM and attention visualisation and emphasise the necessity of embedding transparency directly into detection pipelines rather than treating it as a post-hoc feature.

Haq et al. (2023) introduce a neuro-symbolic multimodal approach to deepfake detection that leverages emotional inconsistencies between facial expressions and vocal tone. By combining deep learning with symbolic reasoning, the framework detects mismatched affective cues that frequently occur in synthetic media. The work demonstrates that emotionally grounded cross-modal reasoning can significantly improve robustness and interpretability.

Qureshi et al. (2024) provide a large-scale survey of multimodal media forensics, reviewing detection techniques for images, video, and audio on social platforms. The study identifies critical challenges, including cross-dataset generalisation, dataset bias, and the lack of unified multimodal benchmarks. The authors conclude that future detectors must operate across modalities to remain effective against evolving deepfake generation techniques.

Wang et al. (2025) propose ERF-BA-TFD+, a state-of-the-art audio-visual deepfake detector combining enhanced convolutional receptive fields with transformer-based temporal modelling. The method achieves top performance on recent audio-visual benchmarks, demonstrating the effectiveness of joint temporal and cross-modal feature modelling.

Finally, Sippy et al. (2024) examine public trust and perception of deepfakes through a large UK-based survey. The findings show high public concern and low confidence in identifying synthetic media, reinforcing the importance of detection systems that prioritise clarity, transparency, and user trust alongside technical accuracy.

2. Skeleton Project Report Structure

2.1 Abstract

The project investigates the development of a context-aware deepfake video detection and verification framework designed to detect AI-generated videos with high robustness and transparency. The proposed system integrates visual, audio, and provenance-based signals using a hybrid CNN-Transformer architecture. Explainable AI mechanisms and metadata verification are incorporated to improve interpretability and user trust. Evaluation is conducted using standard performance metrics and qualitative user feedback.

2.2 Introduction

The rapid advancement of generative AI has enabled the creation of highly realistic deepfake videos, posing severe threats to information integrity, public trust, and digital security. Existing detection methods primarily rely on visual artefacts and often fail when applied to unseen or real-world data.

This project addresses these limitations by proposing a context-aware detection framework capable of reasoning beyond visual features. By incorporating multimodal learning, explainability, and provenance verification, the framework aims to improve robustness, transparency, and real-world applicability.

2.3 Literature Review

Existing deepfake detectors are dominated by CNN-based approaches that perform well on benchmark datasets but generalise poorly. Transformer-based models improve temporal reasoning but introduce interpretability challenges. Recent research demonstrates the effectiveness of multimodal detection by aligning audio and visual signals, yet full integration remains limited.

Explainability remains underdeveloped, with most systems functioning as black boxes. In parallel, provenance-based solutions such as metadata authentication and watermarking show promise but are rarely combined with AI-driven detection.

The literature identifies five persistent gaps: poor generalisation, weak audio-visual fusion, limited explainability, lack of provenance integration, and minimal

user-centric evaluation. The proposed framework is designed to address these gaps in a unified manner.

2.4 Methodology

The framework adopts a hybrid CNN-Transformer architecture. A convolutional backbone extracts spatial features from video frames, while a transformer module models temporal dependencies across frame sequences. Audio features are extracted using MFCCs and aligned temporally with visual data.

Multimodal fusion combines audio and visual representations to detect synchronisation inconsistencies. Explainable AI techniques, including Grad-CAM heatmaps and attention visualisations, are embedded within the inference process. Metadata and provenance checks aligned with C2PA standards complement the detection pipeline.

2.5 System Design

The system is designed as a modular architecture comprising preprocessing, visual analysis, audio analysis, fusion, explainability, and verification modules. Clear separation of components ensures scalability, maintainability, and extensibility.

Data flows from video ingestion through preprocessing and model inference to output generation, which includes prediction scores, visual explanations, and provenance indicators. UML diagrams and architectural schematics are used to formalise the design.

2.6 Implementation Plan

The system is implemented using Python, leveraging PyTorch, OpenCV, and audio processing libraries. Core datasets include DFDC, FakeAVCeleb, and KoDF. Development follows a staged approach: baseline model implementation, multimodal integration, explainability embedding, and system consolidation.

GPU-supported local training is supplemented with cloud resources where required. Model validation occurs iteratively throughout development to minimise overfitting and ensure stability.

2.7 Evaluation Strategy

Quantitative evaluation employs accuracy, precision, recall, F1-score, and AUC across multiple datasets. Robustness testing includes compression and cross-dataset validation.

Qualitative evaluation is conducted through a user study assessing system clarity, trust, and usability using structured questionnaires. The combined evaluation approach ensures both technical effectiveness and user-centred reliability.

2.8 Ethical Considerations

All datasets are either officially licensed or ethically approved. Any user-contributed data is collected with informed consent and anonymised in accordance with GDPR guidelines.

The framework is limited to detection and verification, with no capability for content generation. Bias assessment and transparency are prioritised to minimise ethical and societal risks associated with automated deepfake judgments.

2.9 Future Work and Limitations

Current limitations include dataset dependency and computational constraints. Future work may extend the framework to image-based deepfakes, real-time detection, adversarial robustness, and expanded provenance analysis.

Further improvements may include enhanced user interfaces, larger-scale user studies, and integration with content moderation pipelines.